Exercise 9

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R Lab: Basic Operations of R Programming

#Part A — Creating Data Frames

#Question #1

```
# Part a: create the data frame and save it as shoesize.csv
ShoeSize <- c(6.5, 9.0, 8.5, 8.5, 10.5, 7.0, 9.5, 9.0, 13.0, 7.5, 10.5, 8.5, 12.0, 10.5,
              13.0, 11.5, 8.5, 5.0, 10.0, 6.5, 7.5, 8.5, 10.5, 8.5, 10.5, 11.0, 9.0, 13.0)
Height <- c(66.0, 68.0, 64.5, 65.0, 70.0, 64.0, 70.0, 71.0, 72.0, 64.0, 74.5, 67.0, 71.0, 71.0,
            77.0, 72.0, 59.0, 62.0, 72.0, 66.0, 64.0, 67.0, 73.0, 69.0, 72.0, 70.0, 69.0, 70.0)
Gender <- c("F", "F", "F", "F", "M", "F", "F", "F", "M", "F", "M", "F", "M", "M",
            "M", "M", "F", "F", "M", "F", "F", "M", "M", "F", "M", "M", "M", "M")

shoes_df <- data.frame(ShoeSize = ShoeSize, Height = Height, Gender = Gender)
write.csv(shoes_df, "shoesize.csv", row.names = FALSE)
```

```
# Part b: import the saved csv and display the last 5 rows
shoes <- read.csv("shoesize.csv")
tail(shoes, 5)
```

```
##    ShoeSize Height Gender
## 24      8.5     69      F
## 25     10.5     72      M
## 26     11.0     70      M
## 27      9.0     69      M
## 28     13.0     70      M
```

```
# Part c: structure and content
str(shoes)
```

```
## 'data.frame':    28 obs. of  3 variables:
##  $ ShoeSize: num  6.5 9 8.5 8.5 10.5 7 9.5 9 13 7.5 ...
##  $ Height  : num  66 68 64.5 65 70 64 70 71 72 64 ...
##  $ Gender  : chr  "F" "F" "F" "F" ...
```

```
head(shoes)
```

```
##   ShoeSize Height Gender
## 1      6.5   66.0      F
## 2      9.0   68.0      F
## 3      8.5   64.5      F
## 4      8.5   65.0      F
## 5     10.5   70.0      M
## 6      7.0   64.0      F
```

```r
# Part d: dimensions
dim(shoes)
```

```
## [1] 28  3
```

```r
nrow(shoes)
```

```
## [1] 28
```

```r
ncol(shoes)
```

```
## [1] 3
```

```r
# Part e: summary vs str
summary(shoes)
```

```
##     ShoeSize          Height           Gender
##  Min.   : 5.000   Min.   :59.00   Length   :28
##  1st Qu.: 8.500   1st Qu.:65.75   N.unique : 2
##  Median : 9.000   Median :69.50   N.blank  : 0
##  Mean   : 9.411   Mean   :68.57   Min.nchar: 1
##  3rd Qu.:10.500   3rd Qu.:71.25   Max.nchar: 1
##  Max.   :13.000   Max.   :77.00
```

```r
# Answer: str() shows the structure, like the data type of each column and a
# preview of the values. summary() shows statistics, like the min, median, mean,
# and max for the numeric columns. For Gender it only shows the length and class
# because it is text.
```

```r
# Part f: subsets of males and females with shoe size and height
males <- shoes[shoes$Gender == "M", c("ShoeSize", "Height")]
females <- shoes[shoes$Gender == "F", c("ShoeSize", "Height")]
males
```

```
##    ShoeSize Height
## 5      10.5   70.0
## 9      13.0   72.0
## 11     10.5   74.5
## 13     12.0   71.0
## 14     10.5   71.0
## 15     13.0   77.0
## 16     11.5   72.0
## 19     10.0   72.0
## 22      8.5   67.0
## 23     10.5   73.0
## 25     10.5   72.0
## 26     11.0   70.0
## 27      9.0   69.0
## 28     13.0   70.0
```

```r
females
```

```
##    ShoeSize Height
## 1       6.5   66.0
## 2       9.0   68.0
## 3       8.5   64.5
## 4       8.5   65.0
## 6       7.0   64.0
## 7       9.5   70.0
```

```
## 8       9.0   71.0
## 10      7.5   64.0
## 12      8.5   67.0
## 17      8.5   59.0
## 18      5.0   62.0
## 20      6.5   66.0
## 21      7.5   64.0
## 24      8.5   69.0
```

```
# Answer: There are 14 males and 14 females. The males mostly have bigger shoe
# sizes and are taller than the females.
```

```
# Part g: mean of shoe size and height
mean(shoes$ShoeSize)
```

```
## [1] 9.410714
```

```
mean(shoes$Height)
```

```
## [1] 68.57143
```

```
# Answer: The mean shoe size is about 9.41 and the mean height is about
# 68.57.
```

```
# Part h: relationship between shoe size and height
cor(shoes$ShoeSize, shoes$Height)
```

```
## [1] 0.7766089
```

```
# Answer: Yes, there is a positive relationship. People with bigger shoe sizes
# are usually taller, and the correlation is about 0.78, which is fairly
# strong. This makes sense because taller people usually have bigger feet.
```

#Question #2

```
gender_f <- factor(shoes$Gender, levels = c("F", "M"),
                   labels = c("Female", "Male"))
gender_f
```

```
##  [1] Female Female Female Female Male   Female Female Female Male   Female
## [11] Male   Female Male   Male   Male   Male   Female Female Male   Female
## [21] Female Male   Male   Female Male   Male   Male   Male
## Levels: Female Male
```

```
table(gender_f)
```

```
## gender_f
## Female   Male
##     14     14
```

```
levels(gender_f)
```

```
## [1] "Female" "Male"
```

```
# Answer: Gender is a nominal variable because Female and Male are just
# categories, and one is not worth more than the other. So I made it a factor
# with two levels, Female and Male.
```

#Question #3

```
# Answer: Knitting always uses the folder of the Rmd file as the working
# directory, so R looks for shoesize.csv in the same place every time. If it
# used whatever folder my console was in, the file might not be found on
# another computer. This way, when I share the Rmd and the csv together, it
# runs the same for the other person.
```

#Question #4

```
str(shoes)
```

```
## 'data.frame':    28 obs. of  3 variables:
##  $ ShoeSize: num  6.5 9 8.5 8.5 10.5 7 9.5 9 13 7.5 ...
##  $ Height  : num  66 68 64.5 65 70 64 70 71 72 64 ...
##  $ Gender  : chr  "F" "F" "F" "F" ...
```

```
# Answer: R made Gender a chr (character) column, which is what I expected
# because it contains the letters F and M. ShoeSize and Height are numeric.
# If ShoeSize was imported as character because of a stray value like a
# typo or a letter in the file, then mean() and cor() would not work or give
# NA or errors, and sorting would be wrong because text sorts alphabetically
# (for example, 10.5 would come before 9.0).
```

#Question #5

```
head(shoes)
```

```
##   ShoeSize Height Gender
## 1      6.5   66.0      F
## 2      9.0   68.0      F
## 3      8.5   64.5      F
## 4      8.5   65.0      F
## 5     10.5   70.0      M
## 6      7.0   64.0      F
```

```
tail(shoes)
```

```
##    ShoeSize Height Gender
## 23     10.5     73      M
## 24      8.5     69      F
## 25     10.5     72      M
## 26     11.0     70      M
## 27      9.0     69      M
## 28     13.0     70      M
```

```
# Answer: head() only shows the beginning of the file, so a problem at the
# end would be missed. tail() checks the end. For example, extra blank rows,
# a notes or total row at the bottom, or incomplete rows would only show up
# with tail(). Checking both makes sure the whole file was imported properly.
```

#Question #6

```
# Part a: correlation across all observations
cor(shoes$ShoeSize, shoes$Height)
```

```
## [1] 0.7766089
```

```
# Part b: all rows where ShoeSize is greater than or equal to 11
shoes[shoes$ShoeSize >= 11, ]
```

```
##    ShoeSize Height Gender
## 9      13.0     72      M
## 13     12.0     71      M
## 15     13.0     77      M
## 16     11.5     72      M
## 26     11.0     70      M
## 28     13.0     70      M
```

#Question #7

```
mean(females$Height)
```

```
## [1] 65.67857
```

```
mean(males$Height)
```

```
## [1] 71.46429
```

```
mean(females$ShoeSize)
```

```
## [1] 7.857143
```

```
mean(males$ShoeSize)
```

```
## [1] 10.96429
```

```
# Answer: The females have a mean height of about 65.68 and a mean shoe size
# of about 7.86, while the males have a mean height of about 71.46 and a mean
# shoe size of about 10.96. So the males are taller and have bigger shoes on
# average. The dataset supports a relationship between shoe size and height,
# but gender is also an important factor, because part of the overall
# correlation may come from the males being both taller and having bigger
# shoes. So I should check the relationship separately for each gender.
```